Single mode states → Coherent States or Glauber States

Coherent states are <u>linear</u> superimpositions of number states that represent classical light for $\langle n \rangle \gg 1$ → laser modes EM field above threshold stable amplitude and fixed phase

$$|\alpha\rangle = \exp\left\{-\tfrac{1}{2}|\alpha|^2\right\} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \qquad \alpha = \text{complex number (double continuum)} \quad \{|\alpha|^2 = \langle n \rangle\}$$

$|\alpha\rangle$ form an overcomplete set of <u>non</u> orthogonal nonnormalized states for H.O.s.

$$\langle\alpha|\alpha\rangle = \exp\left(-|\alpha|^2\right) \sum_{n=0}^{\infty} \frac{(\alpha^*)^n \alpha^n}{n!} = \exp\left(-|\alpha|^2\right) \cdot \exp\left(+|\alpha|^2\right) = 1$$

$$\langle\alpha|\beta\rangle = \exp\left(-\tfrac{1}{2}|\alpha|^2 - \tfrac{1}{2}|\beta|^2\right) \sum_{n=0}^{\infty} \frac{(\alpha^*)^n \beta^n}{n!} = \exp\left(-\tfrac{1}{2}|\alpha|^2 - \tfrac{1}{2}|\beta|^2 - \alpha^*\beta\right)$$

$$|\langle\alpha|\beta\rangle|^2 = \exp\left(-|\alpha-\beta|^2\right) \simeq 1 \quad \text{if } |\alpha-\beta| \gg 1$$

Coherent states are eigenstates of $\hat{a}$ [destruction operator] and of $\hat{E}_T^+$ with eigenv. α

$$\hat{a}|\alpha\rangle = \exp\left(-\tfrac{1}{2}|\alpha|^2\right) \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \hat{a}|n\rangle \qquad \left[\hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle\right]$$

$$\hat{a}|\alpha\rangle = \exp\left(-\tfrac{1}{2}|\alpha|^2\right) \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{(n-1)!}} |n-1\rangle = \alpha \cdot \exp\left(-\tfrac{1}{2}|\alpha|^2\right) \sum_{n-1=0}^{\infty} \frac{\alpha^{n-1}}{\sqrt{(n-1)!}} |n-1\rangle = \alpha|\alpha\rangle$$

$\hookrightarrow n=0 \to$ null $\Rightarrow \sum_{n=1}^{\infty} = \sum_{n-1=0}^{\infty}$

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$$

Adjoint equation:

$$\langle\alpha|\hat{a}^+ = \langle\alpha|\alpha^*$$

Expectation value of $\hat{n}$ and $\hat{n}^2$ (1st & 2nd moments)

$$\langle n \rangle = \langle\alpha|\hat{n}|\alpha\rangle = \langle\alpha|a^+a|\alpha\rangle = |\alpha|^2\langle\alpha|\alpha\rangle = |\alpha|^2$$

$$\langle n^2 \rangle = \langle\alpha|\hat{n}^2|\alpha\rangle = \cdots |\alpha|^4 + |\alpha|^2 = \langle n \rangle^2 + \langle n \rangle$$

Variance of n

$$(\Delta n)^2 = \sigma_n^2 = \langle n^2\rangle - \langle n\rangle^2 = \langle n\rangle^2 + \langle n\rangle - \langle n\rangle^2 = \langle n\rangle$$

The variance is equal to the mean value!

Relative variance

$$\frac{(\Delta n)^2}{\langle n\rangle^2} = \frac{\langle n\rangle}{\langle n\rangle^2} = \frac{1}{\langle n\rangle}$$

Probability of finding n photons in the mode in state $|\alpha\rangle$

$$P_n = |\langle n|\alpha\rangle|^2 = \exp(-|\alpha|^2)\frac{|\alpha|^{2n}}{n!} = \exp(-\langle n\rangle)\frac{\langle n\rangle^n}{n!}$$

The poisson distribution describes the probability of finding exactly n photons for a state $|\alpha\rangle$ with $\langle n\rangle$ photons on average $\langle\alpha|\hat{n}|\alpha\rangle = |\alpha|^2 = \langle n\rangle$

Expectation values for $\hat{x}$ and $\hat{y}$ [$|\alpha\rangle$ is right eigenstate of $\hat{a}$ and left eig. of $\hat{a}^+$]

$$\langle\alpha|\hat{x}|\alpha\rangle = \langle\alpha|\frac{\hat{a}^+ + \hat{a}}{2}|\alpha\rangle = \frac{\alpha+\alpha^*}{2} = Re(\alpha) = |\alpha|\cos(\xi) \qquad [\alpha = |\alpha|\exp(i\xi)]$$

$$\langle\alpha|\hat{y}|\alpha\rangle = \langle\alpha|\frac{\hat{a}-\hat{a}^+}{2i}|\alpha\rangle = \frac{\alpha-\alpha^*}{2i} = Im(\alpha) = |\alpha|\sin(\xi)$$

$$\langle\alpha|\hat{x}^2|\alpha\rangle = \langle\alpha|\left(\frac{\hat{a}^+ + \hat{a}}{2}\right)^2|\alpha\rangle = \dots = [Re(\alpha)]^2 + 1/4$$

$$\langle\alpha|\hat{y}^2|\alpha\rangle = \langle\alpha|\left(\frac{\hat{a}-\hat{a}^+}{2i}\right)^2|\alpha\rangle = \dots = [Im(\alpha)]^2 + 1/4$$

$$\left[\begin{aligned}\hat{x}^2 &= 1/4\,(\hat{a}^{+2} + \hat{a}^+\hat{a} + \hat{a}\hat{a}^+ + \hat{a}^2) = 1/4\,(\hat{a}^{+2} + 2\hat{a}^+\hat{a} + \hat{a}^2 + 1)\\ \hat{y}^2 &= 1/4\,(-\hat{a}^{+2} + 2\hat{a}^+\hat{a} - \hat{a}^2 + 1)\end{aligned}\right]$$

$$(\Delta x)^2 = \langle\alpha|\hat{x}^2|\alpha\rangle - \langle\alpha|\hat{x}|\alpha\rangle^2 = [Re(\alpha)]^2 + 1/4 - [Re(\alpha)]^2 = 1/4$$

$$(\Delta y)^2 = \langle\alpha|\hat{y}^2|\alpha\rangle - \langle\alpha|\hat{y}|\alpha\rangle^2 = [Im(\alpha)]^2 + 1/4 - [Im(\alpha)]^2 = 1/4$$

$\Rightarrow (\Delta x)^2(\Delta y)^2 = 1/16$ independently on $\langle n\rangle$

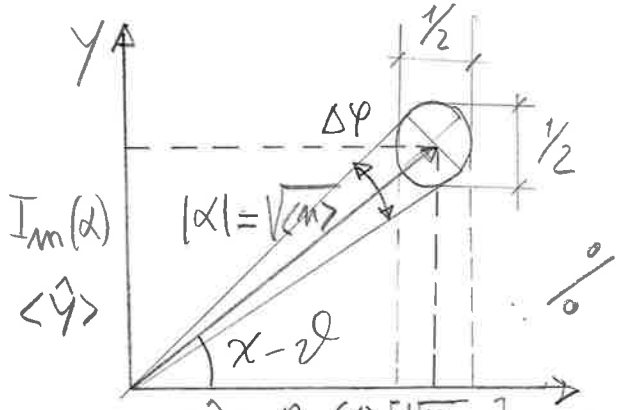
$\hat{x}$ and $\hat{y}$ gives the amplitudes of the cos and sin components of the electric field ($\Delta\phi = \pi/2$): $(\Delta x)^2 = (\Delta y)^2 = 1/4 \;\forall\langle n\rangle$

$\hat{E}_T = \hat{x}\cos\chi + \hat{y}\sin\chi \qquad [\chi = \omega t - kz - \pi/2] \rightarrow$ depends on z

Signal in Homodyne: $\langle\alpha|\hat{E}_T(\chi)|\alpha\rangle = |\alpha|\cos(\chi - \vartheta)$

$$\text{Noise}^2 = \langle\alpha|\hat{E}_T^2(\chi)|\alpha\rangle = \dots = \langle\alpha|\hat{x}^2|\alpha\rangle = \langle\alpha|\hat{y}^2|\alpha\rangle = 1/4$$

$$S^2/4^2 = 4|\alpha|^2\cos^2(\chi-\vartheta) = 4\langle n\rangle\cos^2(\chi - 2\vartheta) \uparrow \text{with } \langle n\rangle$$



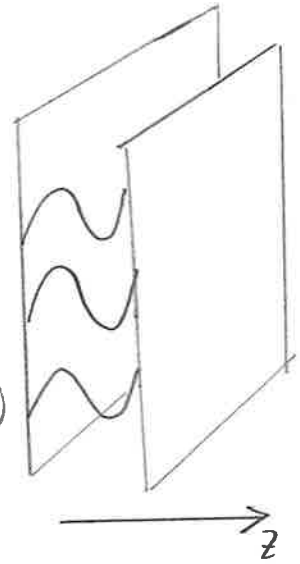
$$\Delta n = \left[\sqrt{\langle n\rangle_{max}}\right]^2 - \left[\sqrt{\langle n\rangle_{min}}\right]^2 = \left[\sqrt{\langle n\rangle} + 1/4\right]^2 - \left[\sqrt{\langle n\rangle} - 1/4\right]^2 = \sqrt{\langle n\rangle} \qquad \Delta\varphi = \frac{1}{2|\alpha|} = \frac{1}{2\sqrt{n}} \qquad [\Delta\varphi\,\Delta n = 1/2]$$

Casimir effect

Hamiltonian of a single mode $\rightarrow k_n$

$$\hat{H}_{k_n,J} = \frac{1}{2}\hbar\omega_n\left(\hat{a}_{k_n,J}\,\hat{a}^{+}_{k_n,J} + \hat{a}_{k_n,J}\,\hat{a}^{+}_{k_n,J}\right) \qquad k_n = \pi n/L \quad \omega_n = c k_n$$

$$\hat{H}_{k_n,J}\,|n_{k_n,J}\rangle = \left(n_{k_n,J} + \frac{1}{2}\right)\hbar\omega_n\,|n_{k_n,J}\rangle$$



The mode corresponds to a plane wave $\Psi(\vec{z},t) = A_n \exp(i\omega_n t)\exp(i k_n z)$

zero point energy $\mathcal{E}_{0n} = \frac{1}{2}\hbar\omega_n$

Considering the infinite set of modes (numerables)

$$\mathcal{E}_0 = \hbar\sum_{n=0}^{\infty}\omega_n = \frac{\pi\hbar c}{L}\sum_{n=0}^{\infty} n$$

When the plates are removed the numerables modes $\rightarrow$ continuum

$$\mathcal{E}_0 = \pi\hbar c\int_0^{\infty}\nu\,d\nu \rightarrow \text{The energy density changes}$$

$$\Delta\mathcal{E}_0 = \mathcal{E}_{cav} - \mathcal{E}_{vac} = \frac{\pi\hbar c}{L}\left(\sum_{n=0}^{\infty} n - \int_0^{\infty}\nu\,d\nu\right) \qquad \infty - \infty$$

Using a convergence factor $\exp(-\varepsilon n)$ $\exp(-\varepsilon\nu)$

$$\sum_{n=0}^{\infty} n e^{-\varepsilon n} = \frac{e^{\varepsilon}}{(e^{\varepsilon}-1)^2} \qquad \int_0^{\infty}\nu e^{-\varepsilon\nu}\,d\nu = 1/\varepsilon^2 \Rightarrow \sum_{n=0}^{\infty} n - \int_0^{\infty}\nu\,d\nu = \lim_{\varepsilon\to 0}\frac{e^{\varepsilon}\varepsilon^2 - e^{2\varepsilon} + 2e^{\varepsilon} - 1}{e^{2}(e^{\varepsilon}-1)^2} = -\frac{1}{12}$$

Then $\Delta\mathcal{E}_0 = -\frac{1}{12}\frac{\pi\hbar c}{L}$ The difference is negative. This mean the removing the plates the energy increases. The Force between the plates is then

$$F_z = -\frac{\partial\mathcal{E}_0}{\partial L} = -\frac{1}{12}\frac{\pi\hbar c}{L^2} \text{ (attractive)}.$$ Using a 3D model $F = -\frac{A\pi^2\hbar c}{240 L^4}$ A = area of the plates

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